

Flagged: Food Label Scanner - Master Product & Technical Brief (V1 MVP)

1. Executive Summary & Strategy

Flagged is a personalized ingredient translator. Health-conscious consumers and allergy parents suffer from "label fatigue" because manufacturers use dozens of confusing chemical names to disguise ingredients.

Instead of forcing users to memorize a chemistry textbook, Flagged uses local on-device OCR to instantly read the physical ingredient paragraph, cross-reference it against the user's personal "Red Flag" settings, and visually highlight the specific ingredients they want to avoid.

- **The Competitive Moat:** Unlike cloud-based competitors (Yuka, Fig) that rely on barcodes and arbitrary "health scores," Flagged is an objective, 100% offline OCR scanner. It reads raw text on any printed label.
 - **Target User:** US moms managing groceries for multi-trigger households (Dye-free/ADHD, severe food allergies/celiac).
 - **The Business Model:** A **Metered Trial (10 Free Scans)** leading to a **One-Time Lifetime Fee**, designed to combat subscription fatigue and prove the technology works before asking for the sale.
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2. Tech Stack & Architecture

To maintain the app's competitive moat of being an "Offline X-Ray," the application requires zero backend architecture. All processing and data storage must happen entirely on-device.

- **Language & UI:** Swift / SwiftUI. (*Android is out of scope for MVP*).
- **Minimum Target OS:** iOS 17.0+ (*Strictly required. The app relies on iOS 17 updates to SwiftData and the Vision framework that are not backwards compatible*).
- **Primary Database (SwiftData):** Used to manage all local persistence, including parsing the bundled JSON dictionary, saving user profiles, storing custom ingredients, and tracking scan metrics.
- **Data Ingestion:** Foundation (Standard `JSONDecoder`). On first launch, the app parses a bundled local JSON file containing the 11 core categories and migrating it into the SwiftData context.
- **The OCR Engine (No Cloud APIs Allowed):**

- *VisionKit (DataScannerViewController)*: Handles the live video feed, text recognition, and projects live UI bounding boxes.
 - *Vision Framework*: Utilizes native item tracking (*RecognizedItem.id*) to ensure a specific block of text isn't processed twice while the camera moves.
 - *Custom Algorithms*: Implement Longest Common Substring (or Levenshtein Distance) to merge trailing/leading text between continuous camera frames, and Regex to sanitize common optical errors.
 - **Monetization Infrastructure: RevenueCat (StoreKit 2).**
 - *Offline Entitlement Handling*: Configured for a **Non-Consumable (One-Time Purchase)**. It must cache the user's "Premium" entitlement locally. If a Premium user opens the app in a store with zero cellular connection, the app must check the cached status via StoreKit 2 to ensure they are not locked out of their app.
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3. Visual Identity & Typography

- **Typography**: Exclusively uses **Atkinson Hyperlegible** to eliminate ambiguous characters (l, I, 1) and reduce eye strain.
 - **Dynamic Color System**: No single hex code is shared between modes.
 - *Dark Mode (Default - X-Ray feel)*: Canvas (*#111827*), Cards/Menus (*#1F2937*), Primary Text (*#F9FAFB*), Muted Text (*#9CA3AF*), Safe/Scanning Cyan (*#22D3EE*), Flagged Red (*#EF4444*).
 - *Light Mode (Clinical feel)*: Canvas (*#F1F5F9*), Cards/Menus (*FFFFFF* with shadow), Primary Text (*#0F172A*), Muted Text (*#475569*), Safe Cyan (*#0E7490*), Flagged Red (*#B91C1C*).
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4. The 6-Screen Onboarding Flow

The logic: Problem → The Tech/Moat → Interactive Setup → Navigation Overview → The Fully Unlocked Trial → Lifetime Upsell Intro.

- **Screen 1: Label Fatigue (The Problem)**
 - *Visual*: A densely packed, confusing ingredient list.
 - *Copy*: "Reading food labels is exhausting. The FDA allows thousands of confusing additives in our food, and memorizing which ones are approved for your family shouldn't be your full-time job."
- **Screen 2: No Barcodes. No Health Scores. (The Moat)**
 - *Visual*: An animation of scanning a handwritten bakery label.

- *Copy*: "Flagged doesn't use barcodes or arbitrary 'health scores.' Our offline X-Ray instantly reads raw text on any printed label."
 - **Screen 3: Personalize Your Scanner (Interactive Baseline Setup)**
 - *Visual*: A clean list of the 11 default filters (e.g., Artificial Dyes, Gluten).
 - *Copy*: "What are you trying to avoid? Select a starting filter to customize your X-Ray (you can change this or add more later)."
 - **Screen 4: The 4-Tab Hub (Navigation Overview)**
 - *Visual*: A sleek graphic highlighting the bottom menu.
 - *Copy*: "Your command center: Home, Scan, Pantry, and Settings."
 - **Screen 5: The 10-Scan Trial (The Hook)**
 - *Visual*: A large graphic of a battery or scanner charging up to "10".
 - *Copy*: "You're all set! You have 10 free scans to test Flagged in the real world. We've unlocked every feature—including Custom Ingredients and Multiple Family Profiles—so you can see the magic for yourself."
 - **Screen 6: The "Founding Member" Transparent Pricing (The Climax)**
 - *Visual*: A clean, high-contrast pricing card.
 - *Copy*: "Ditch the \$40/year subscriptions. After your 10 free scans, unlock unlimited scanning for life for a one-time fee of \$24.99. No hidden fees. Pay once, own it forever."
 - *Action Buttons*:
 - [Start My 10 Free Scans] (Primary)
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5. Define the Data Models (SwiftData Schemas)

5.1. The User & Profile Model

- `profileId`: UUID
- `activeCategories`: Array of Strings (Stores IDs of active default categories).
- `excludedIngredients`: Array of Strings (Stores user-toggled exceptions).
- `customIngredients`: Array of Strings (User-typed strings not found in the JSON).

5.2. The Pantry Item Model (Approved Foods)

- `itemId`: UUID
- `brandName`: String
- `productName`: String
- `imageFilePath`: String (Local URL pointing to the compressed thumbnail.
`.isExcludedFromBackup = true`).
- `originalIngredients`: Array of Strings (An ordered list of the ingredients exactly as they appeared on the package when saved).
- `dateAdded`: Date

- **lastVerifiedDate:** Date (Timestamp of the most recent scan. Used to trigger the 30-day recheck).
- **deletedAt:** Date? (Optional - for the 24-hour Soft Delete window).

5.3. The Lifetime Stats & Free Trial Model (Singleton)

A single, continuously updating record per device.

- **statsId:** UUID
 - **freeScansUsed:** Int (Starts at 0. Caps at 10. Triggers the hard paywall).
 - **totalLabelsRead:** Int (Increments every successful scan).
 - **totalRedFlagsCaught:** Int (Increments by the number of highlighted ingredients found in flagged scans).
 - **totalCleanScans:** Int (Increments by 1 every time a scan results in a Clean Result).
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6. Core Architecture (The 4-Tab Hub)

Standard, 4-tab bottom navigation menu (25% width each). No floating action buttons.

Tab 1: HOME (Dashboard & Filter Hub)

- **The Personalized Header:** "Good Morning, [Name]." (Name is pulled from the Settings tab).
- **"Shopping For" Quick Toggles (Profile Management):** A horizontal row of Profile Chips.
 - *Selecting:* Users tap a family member's chip to activate their specific filters for the current scan.
 - *Editing:* Tapping an [**Edit**] icon on a Profile Chip opens a bottom sheet. This is where the user toggles the 11 default "Red Flag" filters and manages Custom Ingredients *for that specific profile*.
 - *Adding:* All users see a functional [**Add Profile +**] button to freely build multiple profiles.
- **The Protection Summary (The 3 Pillars of Value):**
 - *Labels Read:* Proves sheer volume of text processed and time saved.
 - *Red Flags Caught:* Proves the danger avoided.
 - *Clean Scans:* Proves the positive value/peace of mind delivered.

Tab 2: SCAN (The Core Action Tab)

- **State 1: Standby Mode (Default):**
 - *Top 2/3:* Dark placeholder screen.

- *The Meter (Trial Users Only)*: A prominent pill UI at the top of the screen reading "Scans Remaining: [10 - freeScansUsed] / 10".
- *Bottom 1/3 (Action Menu)*: [Start Camera Scanner] (Primary), [Paste], and [Choose Photo].
- **State 2: The Hard Paywall Lockout:**
 - If `freeScansUsed == 10`, State 1 changes. The Action Menu buttons are replaced by a massive lock icon and a single primary button: [Unlock Unlimited Scans - \$24.99] (Showing the \$39.99 strikethrough). The user can no longer activate the camera, paste text, or upload photos until paid.
- **State 3: Active Mode (Live Scan)**: Live camera feed activates with Cyan bounding box and a 3-second visual countdown.

Tab 3: PANTRY (My Approved List & Audit Hub)

- **The Icon**: A clean, modern pantry jar or open box.
- **Section 1: Skimpflation & Reformulation Checks (The To-Do List)**:
 - *The Trigger*: If any saved item has a `lastVerifiedDate` older than 30 days, it moves out of the main grid and appears here at the top of the screen in a checklist UI.
 - *The Header & Subtitle*: "Skimpflation & Reformulation Checks: Brands sneakily change recipes all the time. Re-scan these items to ensure they are still approved."
 - *The Intercept Modal*: To prevent users from mistakenly scanning the old box still sitting in their house, tapping a task triggers a strict modal: "Only scan a NEWLY PURCHASED box to check for changes. Do you have a new box ready?" -> [Yes, open camera] or [Remind me later].
- **Section 2: My Safe Foods (The Grid)**: A scrollable grid displaying the user's saved, up-to-date safe foods.
- **Section 3: The "Recent Changes" Log (24-Hour Undo)**: At the very bottom, a list view displaying a 24-hour history of deleted items with an [Undo] button. Once 24 hours pass, items are permanently purged.
- **The Empty State**: If no items exist in the grid or the log, display a visually calming graphic with copy: "Your safe list is empty. When you scan an item with no red flags, save it here so you never have to second-guess it again."

Tab 4: SETTINGS (App Administration)

Strictly for global app management and Apple compliance.

- **User Details**: A text field to enter "First Name".
- **Purchases**: A [Restore Purchases] button and a status indicator.
- **Support**: [Report an Issue / Contact Us] button.
- **Legal**: [Privacy Policy] and [Terms of Service] links.

7. The OCR Engine: Capture & Translation

The user does not press a shutter button. They point the camera, and the app does the rest.

Step 1: The 3-Second Live Scan (Frame Processing)

- **Dynamic Highlighting:** Bounding boxes (Cyan #22D3EE stroke, transparent fill) project over paragraphs.
- **Deduplication & Spatial Sorting:** Extract text, prevent duplicate block reads via `RecognizedItem.id`, and sort based on X/Y coordinates to rebuild the paragraph.
- **String Merging Algorithm:** Implement Longest Common Substring/Levenshtein to stitch trailing text of Frame 1 with leading text of Frame 2.

Step 2: Quality Control & Normalization

At exactly 0 seconds, stitching stops.

- *Validation:* App checks for start indicators (e.g., "Ingredients:"). If illegible, abort and show error (*this does NOT consume a free scan*).
- *Normalization:* Lowercase text, strip line-break hyphens, tokenize via commas/parentheses.

Step 3: The Matching Algorithm (Hybrid Approach)

1. **Regex Cleaning:** Sanitize OCR errors (0 to 0, 1 to 1).
2. **Exact Match:** Check tokens against the SwiftData dictionary (and Custom Ingredients assigned to the active Profile) for a 1:1 match.
3. **Fuzzy Match:** Apply Levenshtein distance. If a token has an 85%+ similarity to a red-flag term, it triggers a flag.

8. The Results Screen (Static Output)

Outcome A: The Clean Result

- **Header:** Cyan icon: "No red flags detected."
- **The Ingredient List:** Full paragraph displayed in plain white text.
- **Primary Action Button:** [Save to Pantry]. Tapping this opens a quick camera viewfinder asking the user to "Snap a photo of the front of the packaging." Once snapped, a modal prompts for **Brand Name** and **Product Name** to save it to their Pantry tab.

Outcome B: The Flagged Result

- **Header:** Alert Red icon: "Red flags detected."
- **The Ingredient List:** Full paragraph displayed. Bad ingredients are highlighted in **Alert Red**.
- **The Breakdown:** Charcoal card explains exactly *why* it was flagged.

Secondary Action Buttons (Both Outcomes):

- [Scan Another Item]. (If free user just used their 10th scan, this button changes to [Unlock Unlimited Scans]).
- [Return to Home].

8.1: The Rescan Engine (Pantry Lite Comparison)

8.1. The Pantry Recheck Flow (Lite Diff Engine + Red Flag Check)

When a user scans a new box from the Pantry's "Recheck" section, the app performs a cross-reference between the newly scanned ingredients and the `originalIngredients` array saved in SwiftData.

The Three Background Checks (Evaluated instantly upon scan):

1. **The Reformulation Check:** Checks if any ingredients are missing from the old list, or if new ingredients were added.
2. **The Skimpflation Check:** Checks if the relative order of the remaining ingredients shifted.
3. **The Red Flag Check:** If *any* change is detected in steps 1 or 2, the app automatically runs the new ingredient array through the user's active Red Flag filters.

The Recheck Results UI:

- **Outcome 1: Identical & Safe**
 - Show a green success toast ("No changes detected.").
 - Reset the `lastVerifiedDate` to today, and return the item to the normal Pantry grid.
- **Outcome 2: Recipe Changed, but STILL APPROVED**
 - The app pauses and presents a Yellow/Orange "Recipe Change Detected" screen.
 - *The Breakdown:* It plainly highlights the Skimpflation (order shift) or Reformulation (additions/deletions), but displays a clear badge reading: **"No Active Red Flags Detected."**
 - *The Choice:* Ask the user: "The recipe changed, but we didn't catch any of your active red flags. Do you want to keep this item in your pantry?" -> [Keep Item] or [Delete Item].

- **Outcome 3: Recipe Changed AND RED FLAGS CAUGHT (The Climax)**
 - The app triggers the massive **Alert Red** warning screen.
 - *The Breakdown:* It highlights the specific new ingredients that were added (e.g., "Added: Red 40, High Fructose Corn Syrup") and **explicitly shows which active filter caught it (e.g., "Matches Tommy's Artificial Dyes filter")**.
 - *The Choice:* "Red Flag Detected! This new recipe contains an ingredient you are avoiding. Do you want to remove this item from your pantry?" -> [**Delete Item**] (Primary) or [**Keep Item**] (Muted Secondary).

(Developer Note: If the user taps [**Keep Item**] on Outcome 2 or 3, the app updates the *originalIngredients* array to the new baseline and resets the 30-day timer. If they tap [**Delete Item**], it is sent to the "Recent Changes" log with the 24-hour undo timer).

9. Monetization Strategy: The \$24.99 Lifetime Unlock

The app strictly avoids subscriptions and feature-gating. Conversion is driven entirely by a fully unrestricted trial of the technology, followed by a hard usage gate.

1. **The Fully Unlocked Metered Trial:** Free users get exactly **10 successful scans**. During these 10 scans, they have 100% unrestricted access to the entire app. They can create Multiple Profiles, add Custom Ingredients, and build out their Pantry. (Developer Note: A scan is *ONLY* counted toward *freeScansUsed* if it successfully extracts text and routes to the Results Screen).
2. **The Core Feature Lockout:** On the 11th attempt, the scanning engine is entirely locked. The user has now fully experienced the value of the custom profiles, the Safe List, and the OCR. To continue checking food, they must pay the \$24.99 one-time fee.
3. **The Pitch:** "Ditch the \$40/year subscriptions. Get unlimited, offline label reading for life for less than the cost of one year of the other guys."

10. App Store Review Strategy (ASO)

(Uses native *SKStoreReviewController* API)

1. **The "Aha!" Trigger:** Fires immediately after the user successfully scans and catches their *fifth* flagged ingredient during their 10 free scans. It should not trigger until the user has left the results screen.
2. **The "Habit" Trigger:** Fires on the first app open after 30 days of active Premium installation, or upon hitting 50 total scans.